* Edge Detection

→ Edge are local feature of the image that capture important information related to its semantic content.

* 1D step-Edge

→ sharp change of 1D signal

→ In the transition region between the two constant level the abs valve of the first derivative gets high.

→ By fixing a threshold, an edge can be detected.

* 2D step-edge

→ Characterized not only by its strength (magnitude) but also it's direction.

$$\nabla I(x,y) = \frac{\partial I(x,y)}{\partial x} i + \frac{\partial I(x,y)}{\partial y} j$$

→ The gradient is a vector composed by the derivatives along horizontal and vertical axis.

→ Gradient direction gives the direction along which the function exhibits its maximum variation.

* Discrete Gradient Approximation

→ Approximation of partial derivatives as a difference.

* Backward:

$$\partial_x I(i,j) \approx I(i,j) - I(i,j-1)$$

$$\partial_y I(i,j) \approx I(i,j) - I(i,j-1)$$

* Forward

$$\partial_x I(i,j) \approx I(i,j+1) - I(i,j)$$

$$\partial_y I(i,j) \approx I(i,j+1) - I(i,j)$$

→ Forward & backward diff are equivalent to applying two cross-correlation with kernels $\begin{pmatrix} -1 & 1 \end{pmatrix}$ & $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$.

* Central Difference

$$\partial_x I(i,j) \approx I(i,j+1) - I(i,j-1)$$

$$\partial_y I(i,j) \approx I(i,j+1) - I(i,j-1)$$

→ Equivalent two-correlations with kernels $\begin{pmatrix} -1 & 0 & 1 \end{pmatrix}$ & $\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$

* Discrete magnitude approximation

→ the gradient magnitude can be approximated in different ways by using the approximated partial derivates:

* $\|\nabla I\| = \sqrt{(\partial_x I)^2 + (\partial_y I)^2}$

* $\|\nabla I\|_+ = |\partial_x I| + |\partial_y I|$

* $\|\nabla I\|_{max} = \max(|\partial_x I|, |\partial_y I|)$

- Among all, $\|\nabla\|_{max}$ is the most isotrophic (gives a more consistent response).

→ In practice, the signal of an image is not always smooth due to noise.

→ Derivatives amplify noise and are therefore unable to recognize edges.

→ Smoothing the signal before computing the derivatives allows to reduce the noise but also blurs the edges, making it more difficult to localize them.

→ A solution is to smooth & differentiate in a single operation by approximating the gradient as a difference of averages.